Rahul Chajwa

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RESEARCH INTERESTS: Soft Condensed Matter, Biological Physics, Ocean Carbon Cycle

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ACADEMIC POSITION
Mar 2021 - Present

Postdoctoral Researcher, Department of Bioengineering, **Stanford University**
former Human Frontier Science Program Cross Disciplinary Fellow
Advisor: Prof. Manu Prakash, manup@stanford.edu

EDUCATION

- **PhD in Physics**
  May 2015 - Feb 2021

  International Centre for Theoretical Sciences, TIFR Bangalore, India
  Advisors: Prof. Sriram Ramaswamy, FRS, sriram@iisc.ac.in,
  Prof. Rama Govindarajan, rama@icts.res.in,
  Prof. Narayanan Menon, menon@physics.umass.edu

  Mar 2017 – Sep 2017

  University of Massachusetts Amherst, Short-term Scholar
  Advisor: Prof. Narayanan Menon, Department of Physics

- **BS-MS in Physics**
  Aug 2010 - May 2015

  Indian Institute of Science Education and Research, Mohali, India

- **Other Certifications**

  Machine Learning Specialization**,** Coursera
  DeepLearning.AI, Stanford University

  Stanford IGNITE Full Time 2025, Business Fundamentals and Entrepreneurship Training, Stanford Graduate School of Business.

RESEARCH

- **Postdoctoral Research** at Stanford University — Mar 2021 – Present

  Projects: **1) Biophysics of Marine Snow: real-time observation of a sinking microbial ecosystem.** Combining table-top experiments conducted at sea and theory to study biotic aggregates in the ocean and the bacterial dynamics around them. (Published in ***Science*** and another manuscript nearing submission)

  2) Inflation induced motility of a non-motile plankton, to study how plankton cells in the ocean regulate buoyancy and the constrain imposed by gravity on cytoplasmic and vacuolar density. (published in ***Current Biology***)

  Ocean Expeditions: participated in the following research expeditions to study the microscale physics underling the biological carbon pump:

  2025 06/24 – 07/10 in the **Atlantic Ocean** aboard the research vessel **R/V Neil Armstrong** as part of project on Virus Infection at the Interface of Sinking, Turbulence, and Aggregation (VIISTA2).

  2024 05/28 – 06/10 in the **Pacific Ocean** aboard the research vessel **R/V Kilo Moana** as part of project on Virus Infection at the Interface of Sinking, Turbulence, and Aggregation (VIISTA1).

2023 05/17 – 05/25 in the **Pacific Ocean** aboard the research vessel **R/V Sally Ride** as part of Research at the Interface of Phytoplankton, Particles and Lipid Export (RIPPLE2) project.

2022 06/02 – 07/02 in the **Pacific Ocean** aboard the research vessel **R/V Sikuliaq** as part of the Iron and Ocean Acidification (FeOA) project.

2021 06/10 – 06/22 in the **Atlantic Ocean** aboard the research vessel **R/V Endeavor** as part of Research at the Interface of Phytoplankton, Particles and Lipid Export (RIPPLE1) project.

- **PhD Research** at Tata Institute of Fundamental Research (TIFR). Aug 2015 – Feb 2021
  Thesis Title: **Driven Stokesian Suspensions: particle anisotropy, effective inertia and transient growth**
  My PhD research was conferred the **TIFR Best Thesis award in Physics 2021**, with the following citation:

  "For comprehensive work on Stokesian sedimentation including experimental observations of rare new singularities, and theoretical (analytical and numerical) discovery of emergent Hamiltonians, Keplerian orbits, and growing perturbations in linearly stable systems".

- **MS Thesis Research** at Indian Institute of Science Education and Research Aug 2014 – May 2015
  Thesis Title: **Instabilities in Sedimenting Suspensions:** Constructed broken-symmetry hydrodynamics for lattice of sedimenting suspensions with shape anisotropy and conducted their stability analysis.

- **Undergraduate Research**
  a) Indian Institute of Science, Summer Research Fellowship with Prof. Sriram Ramaswamy at Tata Institute of Fundamental Research Hyderabad**.** *Project:* **Stokesian sedimentation of chiral objects** May - July 2013
  b) Summer Training in Physics, at Indira Gandhi Centre for Atomic Research, Kalpakkam with Dr. Baskaran. *Project:* **Fabricating and characterizing superconducting thin films for quantum interference devices.** May - June 2012
  c) Summer Research at Indian Institute of Science Education and Research Mohali with Prof. Jasjeet Bagla, Astrophysics. *Project:* **Gravitational Simulations of planetary orbits and clusters.** May - July 2011

TEACHING EXPERIENCE

- **TA for Physiology Course at the Marine Biological Laboratory, 2024**
  with Prof. Manu Prakash
  Two weeks long course in which I mentored three grad students on a project on the experimental and theoretical aspects of marine snow in coastal ecosystems.

- **TA for Statistical Mechanics at TIFR Hyderabad, 2016**
  with Prof. Mustansir Barma
  Mandatory PhD level Physics Course carrying 4 credits
  My duties involved making and evaluating assignment and exam questions; and holding tutorials.

- **TA for SageMath Programming at IISER Mohali, 2014**
  with Prof. Kapil Paranjape, Mathematics Department
  Mandatory Undergraduate Semester-long course carrying 4 credits
  My duties involved organizing hands-on programming sessions twice a week.

- Working towards the Postdoctoral Teaching Certificate at Stanford University.

MENTORING EXPERIENCE

TA for Physiology Course at the Marine Biological Laboratory, 2024 with Prof. Manu Prakash. Two weeks long course in which I mentored 3 grad students (Anissa Dea, Sajjad Norouzi, Debraj Ghose). 1 grad student (Harshit Joshi with Prof. Rama Govindarajan), 3 undergraduates (Sebastian Burger and Farid Zaheer with Prof. Rama Govindarajan and Rintaro Kirikawa with Prof. Narayanan Menon). This included mentoring through the DAAD RISE program, for which I submitted the proposal.

OUTREACH AND SERVICE

- Presented in the **Pint of Science**, Palo Alto May 2025. Title: *Sinking Snacks and Ocean Hacks*.
- Organized Magnifying Science in Field Settings outreach event as part of **Promise in Science and Mathematics (PRISM)** at the International Center for Theoretical Sciences TIFR, May 2024.
- Organized a Squishy Science at Sea event as part of the **American Physical Society (DSOFT) Squishy Science Sunday** outreach at the APS March Meeting 2024.
- Teaching in the **Building UP Developing Scientist (BioBUDS)** at Stanford University, Mar 2024.
- Teaching in **Stanford Splash** Fall 2022.
- Served as a jury for science exhibits in the **National Children Science Congress** 2019, a program by Department of Science & Technology, Government of India.

HONOURS & AWARDS

- **Invited Panelist with the Science and Technology Advisor to the Secretary of State** This was a high-level panel to discuss the lights and shadows of multilateral scientific collaboration at the 23rd HFSP Awardees Meeting, Washington DC (2024).
- **Stanford Bio-X Travel Award**, Stanford University (2023). **500 USD** for APS March Meeting travel 2023.
- Selected for Participation in the **10th Global Young Scientists Summit** organized by the National Research Foundation, Singapore (2022).
- **HFSP Cross Disciplinary Fellowship**, International Human Frontier Science Program Organization (2021).
  Award amount: **203000 USD** for the period 1 May 2021 – 30 April 2024.
  HFSPO further highlighted me in their annual report 2021.
- **TIFR Best Thesis Award in Physics 2021**, Tata Institute of Fundamental Research, Mumbai. This is a highly competitive award that selects one PhD thesis across all TIFR centers.
- **Infosys Foundation ICTS Excellence Grant**, International Centre for Theoretical Sciences TIFR (2020)
- **IAS Summer Research Fellowship**, Indian Academy of Sciences, Bangalore (2013)
- **Innovation in the Pursuit of Scientific Excellence (INSPIRE) Scholarship** (2010 - *5 years*) awarded by the Department of Science and Technology, India to meritorious students with aggregate marks within the top 1% of the Class XII Board Exam, upon qualifying IISER Aptitude Test.
- **Union Public Service Commission, NDA & NA exam** (2010). Selected for Flying Branch in the Air Force, with all India rank 83 among ~300 thousand applicants.

INVITED TALKS

- Invited contributor of the **Living Futures** Essay: *Towards a Physics of the Living Ocean*. This is a physics community effort to complement and contrast the talks in Living Histories.
- Cellular Slingshots and Hidden Comet-tails in the Ocean's Biological Pump, CDR Coffee Talk, Yale Centre for Natural Carbon Capture, Earth & Planetary Sciences, **Yale University** 2025.
- Suspensions with Internal Degrees of Freedom, Physics Colloquium, **University of Oregon**, 2025.
- Cellular Slingshots and Hidden Comet-tails in the Ocean, **SLAAM (Soft, Living, Active and Adaptive Matter) Seminar**, American Physical Society (DBIO, DSOFT) 2025.
- Suspensions with Internal Degrees of Freedom, James Franck Institute, **University of Chicago** 2025.
- Systems Biology Seminar, **Institute for Advanced Study, Princeton** 2024.
- Flow and Self-Organization in the Biological Pump, **Kavli Institute for Theoretical Physics** 2024.
- Hidden Comet-tails in Marine Snow Impede Ocean-based Carbon Sequestration, International Human Frontier Science Program Organization Meeting 2024, **National Academy of Sciences**, **Washington DC**.
- Surprises in Driven Stokesian Suspensions: on land and sea, Physics Seminar, **Indian Institute of Technology Gandhinagar** 2024
- Cellular Slingshots and Hidden Comet-Tails in the Oceans, Fluid and Turbulence Seminar, **International Center for Theoretical Sciences TIFR** 2024
- Surprises in Driven Stokesian Suspensions: on land and sea, Physics Seminar, **Tata Institute of Fundamental Research Hyderabad** 2024.
- Surprises in Driven Stokesian Suspensions, Soft Matter Seminar, **Indian Institute of Science Education and Research Mohali** 2024.
- Active Caustics, Active matter in complex environments conference, **Aspen Center for Physics** 2023
- Motile particles without inertia form caustics in vortical flows, **APS March Meeting 2022**.

PUBLICATIONS AND PREPRINTS

[8] **R. Chajwa**, Q. Zhang, A. Sharma, M. Prakash, *Dynamics of Bacterial Hives in the Ocean* (2025). [To be submitted]

[7] H. Joshi, **R. Chajwa**, S. Ramaswamy, N. Menon, R. Govindarajan. *Dynamics and clustering of sedimenting disc lattices. **Journal of Fluid Mechanics*** 1017:A1. doi:10.1017/jfm.2025.10467 (2025).

[6] **R. Chajwa**, E. Flaum, K.D. Bidle, Benjamin Van Mooy, Manu Prakash, *Hidden Comet-Tails of Marine Snow Impede Ocean-based Carbon Sequestration, **Science*** 386, eadl5767 (2024). 10.1126/science.adl5767

Perspective: B. B. Cael, Lionel Guidi, *Tiny comets under the sea*. ***Science*** 386, 149-150(2024). 10.1126/science.ads5642

Featured in: ScienceNews, Stanford Report, New York Times

[5] **R. Chajwa**, Rajarshi, Rama Govindarajan, Sriram Ramaswamy, *Active Caustics*, arXiv:2310.01829 [cond-mat.soft](2024) [Under Review]

[4] A.G. Larson*, **R. Chajwa***, Hongquan Li, Manu Prakash, *Inflation induced motility for long distance vertical migration*, **Current Biology** 34, 1-15 (2024). * Equal Contribution

Perspective: Clotilde Cadart, *Cell Biology: Wanderers that balloon towards light*, Current Biology vol. 34, 22 (2024)

Featured in: ***Nature*** 634, 1021 (2024), ScienceNews, BBC

[3] **R. Chajwa**, N. Menon, S. Ramaswamy, R. Govindarajan, *Waves, Algebraic Growth, and Clumping in Sedimenting Disk Arrays*, **Phys. Rev. X** 10, 041016 (2020).

[2] L.P. Dadhichi, J. Kethapelli, **R. Chajwa**, A. Maitra and S. Ramaswamy, *Nonmutual torques and the unimportance of motility for long-range order in two-dimensional flocks*, **Phys. Rev. E** 101, 052601 (2020).

[1] **R. Chajwa**, N. Menon and S. Ramaswamy, *Kepler Orbits in Pairs of Disks Settling in a Viscous Fluid*, **Phys. Rev. Lett.** 122, 224501 (2019).

CONTACT REFERENCES

	RELATIONSHIP
[1] Prof. **Sriram Ramaswamy**, (sriram@iisc.ac.in) Department of Physics, Indian Institute of Science.	PhD Advisor
[2] Prof. **Manu Prakash**, (manup@stanford.edu) Department of Bioengineering, Stanford University.	Postdoctoral Advisor
[3] Prof. **Narayanan Menon,** (menon@physics.umass.edu) Department of Physics, University of Massachusetts Amherst.	PhD Advisor
[4] Prof. **Rama Govindarajan**, (rama@icts.res.in), International Centre for Theoretical Sciences TIFR.	PhD Advisor
[5] Prof. **Kay D. Bidle**, (bidle@marine.rutgers.edu) Department of Marine and Coastal Sciences, Rutgers University.	Postdoctoral Collaborator

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